

Matematisk Statistik

The Bootstrap

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The Bootstrap



pull oneself up by one's bootstraps:
to help oneself without the aid of others; use one's resources
(www.collinsdictionary.com)

Grundlæggende problemstilling: Vi står med data $x_1, \dots, x_n$ og en stikprøvefunktion $g(x_1, \dots, x_n)$, f.eks. gennemsnittet:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Vi kender $g(x_1, \dots, x_n)$, men ikke $g(X_1, \dots, X_n)$ (stikprøvefordeling/sampling distribution) der afhænger af F .

Idé: Betragt den fordeling der opstår, hvis vi trækker fra $x_1, \dots, x_n$ med *tilbagelægning*.

Notation: $x_1^*, \dots, x_n^*$.

The Bootstrap Idea

The original sample approximates the population from which it was drawn. So resamples from this sample approximate what we would get if we took many samples from the population. The bootstrap distribution of a statistic, based on many resamples, approximates the sampling distribution of the statistic, based on many samples.

The Bootstrap

Table 5.1 All possible samples of size 3 from 3969, 3204, and 2892.

x_1^*	x_2^*	x_3^*	$\bar{x}^*$
3969	3969	3969	3969
3969	3969	3204	3714
3969	3969	2892	3610
3969	3204	3969	3714
3969	3204	3204	3459
3969	3204	2892	3355
3969	2892	3969	3610
3969	2892	3204	3355
3969	2892	2892	3251
3204	3969	3969	3714
3204	3969	3204	3459
3204	3969	2892	3355
3204	3204	3969	3459
3204	3204	3204	3204
3204	3204	2892	3100
3204	2892	3969	3355
3204	2892	3204	3100
3204	2892	2892	2996
2892	3969	3969	3610
2892	3969	3204	3355
2892	3969	2892	3251
2892	3204	3969	3355
2892	3204	3204	3100
2892	3204	2892	2996
2892	2892	3969	3251
2892	2892	3204	2996
2892	2892	2892	2892

Fordelingen svarende til $x_1^*, \dots, x_n^*$ kaldes den *teoretiske bootstrap fordeling*.

For $n = 3$ er der $3^3 = 27$ måder at trække med tilbagelægning (når vi tager højde for rækkefølge). Dog kun $\binom{5}{2} = 10$ forskellige værdier af $\bar{x}^*$.

The Bootstrap

I princippet kunne vi godt beregne sandsynlighederne i bootstrap fordelingen, men i praksis er det lettere at simulere udfald fra den.

Bootstrap for a Single Population

Given a sample of size n from a population:

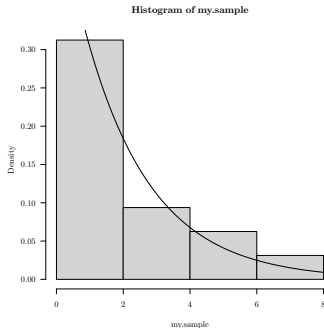
- 1) Draw a resample of size n with replacement from the sample. Compute a statistic that describes the sample, such as the sample mean.
- 2) Repeat this resampling process many times, say, 10 000.
- 3) Construct the bootstrap distribution of the statistic. Inspect its center, spread, and shape.

Vi kan let trække fra bootstrapfordelingen ved at bruge **sample** kommandoen.

Eksempel

```
set.seed(123)
my.sample <- rgamma(16, 1, 1/2)
mean(my.sample)
sd(my.sample)
```

```
> mean(my.sample)
[1] 2.322472
> sd(my.sample)
[1] 2.183536
```



Eksempel

```
N <- 10^5
my.boot <- numeric(N)
for (i in 1:N) {
  x <- sample(my.sample, 16, replace = TRUE) # draw resample
  my.boot[i] <- mean(x) # compute mean, store in my.boot
}
mean(my.boot)
sd(my.boot)
```

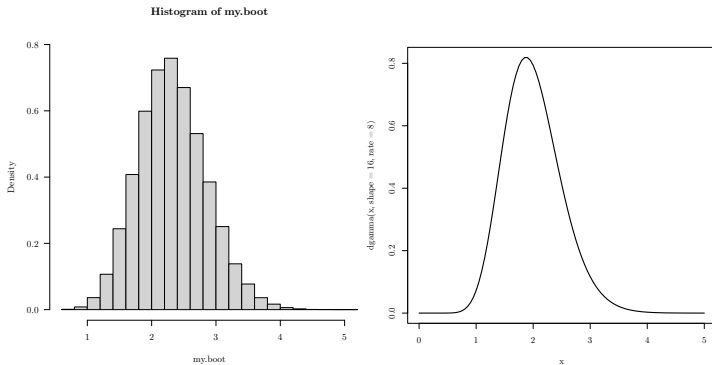
```
> mean(my.boot)
[1] 2.32181
> sd(my.boot)
[1] 0.5298087
```

Eller vi at bruge **replicate** :

```
my.boot <- replicate(N, mean(sample(my.sample, replace = TRUE)))
mean(my.boot)
sd(my.boot)
```

```
> mean(my.boot)
[1] 2.32338
> sd(my.boot)
[1] 0.5277297
```

Eksempel

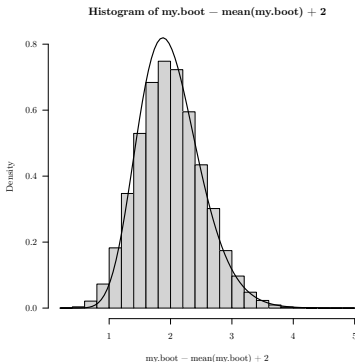
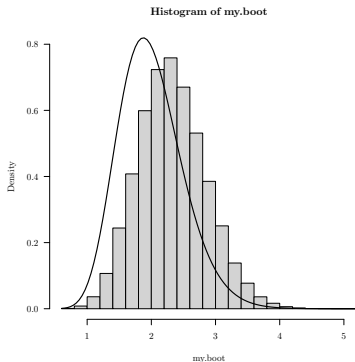


Bootstrap Distributions and Sampling Distributions

For most statistics, bootstrap distributions approximate the spread, bias, and shape of the actual sampling distribution.

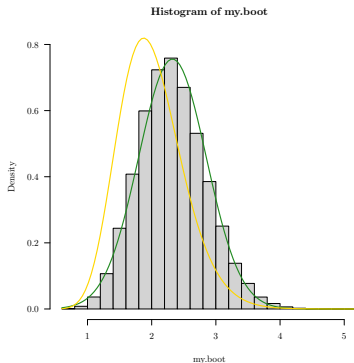
Bemærk: “bias” *ikke* “center”.

Eksempel



Til venstre: Bootstrapfordelingen sammen med stikprøvefordelingen.
Til højre: Bootstrapfordelingen centreret i 2 sammen med stikprøvefordelingen.

Eksempel



Bootstrappedfordelingen sammenlignet med normalapproximationen (grøn) og stikprøvefordelingen (gul).

The Bootstrap kan opfattes som et eksempel på plug-in principle, hvor vi erstatter den ukendte fordelingsfunktion F med den *empiriske fordelingsfunktion* $\hat{F}_n$.

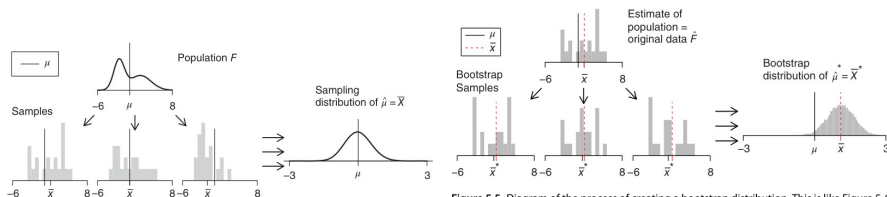


Figure 5.4 Diagram of the process of creating a sampling distribution. Many (infinitely many) samples are drawn from the population, a statistic like $\bar{X}$ is calculated for each. The distribution of the statistics is the sampling distribution.

Figure 5.5 Diagram of the process of creating a bootstrap distribution. This is like Figure 5.4, except that the original data take the place of the population. We draw many samples from the original data, calculate $\bar{X}^*$ or another statistic for each, and collect the statistics to form the bootstrap distribution.

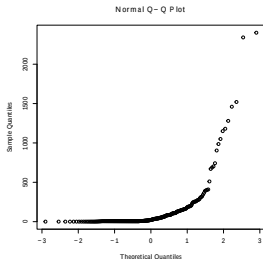
“For most common estimators and under fairly general distribution assumptions, the following hold.”

- *Spredning/“spread”* ✓ Spredningen af bootstrapfordelingen svarer til spredningen af stikprøvefordelingen - $SE[\hat{\theta}^*] \approx SE[\hat{\theta}]$
- *Bias* ✓ Bias af bootstrapfordelingen svarer til bias af stikprøvefordelingen - $\text{Bias } \hat{\theta}^* = \mathbb{E}[\hat{\theta}^*] - \hat{\theta} \approx \mathbb{E}[\hat{\theta}] - \theta = \text{Bias } \hat{\theta}$
- *Skævhed* ✓ Skævhed af bootstrapfordelingen svarer til skævhed af stikprøvefordelingen
- *Center/position* ✗ Bootstrapfordelingen vil være centreret samme sted som den oprindelige stikprøve, f.eks. vil bootstrap fordelingen $\bar{X}^*$ være centreret omkring $\bar{x}$. (og ikke μ).

Bemærk til sidste punkt: Man skal *ikke* forvente at bootstrap-metoden skal kunne bruges til at opnå forbedrede estimater.

Example 5.3

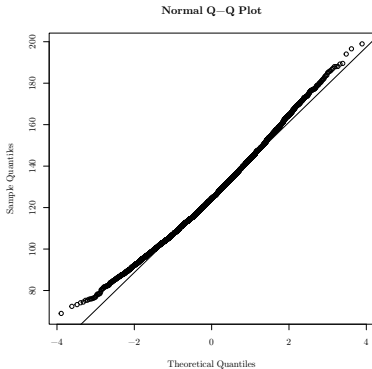
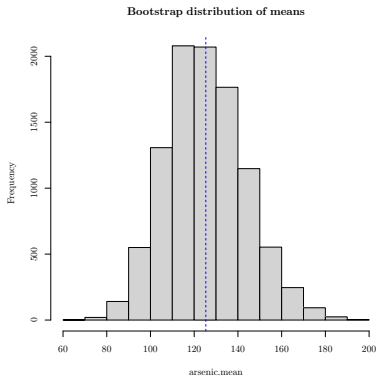
Eksemplet vedrører målinger af arsenik i vandmålinger i Bangladesh. Vi undersøger bootstrap fordelingen af gennemsnittet.



```
library(resampledData3)
Arsenic <- Bangladesh$Arsenic
n <- length(Arsenic)
N <- 10^4
arsenic.mean <- replicate(N,mean(sample(Arsenic, replace = TRUE)))

hist(arsenic.mean, main = "Bootstrap distribution of means")
abline(v = mean(Arsenic), col = "blue", lty = 2)
```

Example 5.3



```
> mean(arsenic.mean) # bootstrap mean
[1] 125.3243
> mean(arsenic.mean)-mean(Arsenic) # bias
[1] 0.00437369
> sd(arsenic.mean) # bootstrap SE
[1] 18.15877
```

Example 5.3

```
> mean(arsenic.mean)-1.96*sd(arsenic.mean)
[1] 89.73312
> mean(arsenic.mean)+1.96*sd(arsenic.mean)
[1] 160.9155
> sum(arsenic.mean > 160.9155)/N
[1] 0.0333
> sum(arsenic.mean < 89.73312)/N
[1] 0.0159
```

Vi ser at cirka 1.6% af bootstrap samples er mindre end gennemsnit minus $2 * SE$ og 3.3% er over gennemsnit plus $2 * SE$, så vi konkluderer at stikprøvefordelingen for gennemsnittet ikke kan betragtes som normalfordelt.

Bootstrap Percentile Confidence Intervals

The interval between the 2.5 and 97.5 percentiles of the bootstrap distribution of a statistic is a 95% *bootstrap percentile confidence interval* for the corresponding parameter.

Beregnes ved hjælp af `quantile(x, probs=c(0.025, 0.975))`, hvor `x` indeholder bootstrap samples.

For arsenik-eksemplet finder vi

```
quantile(arsenic.mean, probs=c(0.025, 0.975))
```

som giver

2.5%	97.5%
92.80887	163.96348

Bemærk at dette kan skrives som $(\bar{x}^* - 32.5154, \bar{x}^* + 38.63918)$ - ikke symmetrisk omkring $\bar{x}^*$.

Percentil Intervallerne skal betragtes med visse forbehold - kan være overraskende dårlige.

Bootstrap for Comparing Two Populations

Given independent samples of sizes m and n from two populations:

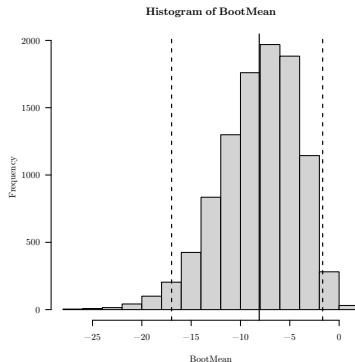
- 1) Draw a resample of size m with replacement from the first sample and a separate resample of size n from the second sample. Compute a statistic that compares the two groups, such as the difference between the two sample means.
- 2) Repeat this resampling process many times, say, 10 000.
- 3) Construct the bootstrap distribution of the statistic. Inspect its spread, bias, and shape.

```
ILEC <- subset(Verizon, select = Time, subset = Group=="ILEC", drop=TRUE)
CLEC <- subset(Verizon, select = Time, subset = Group=="CLEC", drop=TRUE)
m <- length(ILEC)
n <- length(CLEC)
B <- 10^4

BootMean <- numeric(B)

for(i in 1:B) {
  sampleILEC <- sample(ILEC, m, replace=TRUE) # "m" er strengt tage ikke nødvendig.
  sampleCLEC <- sample(CLEC, n, replace=TRUE)
  BootMean[i] <- mean(sampleILEC) - mean(sampleCLEC)
}
```

replicate kan også bruges.



Histogram af bootstrapfordelingen for gennemsnittet. De stiplede linjer er empiriske 2.5% og 97.5% fraktiler.

```
quantile(BootMean, c(0.025, 0.975))
```

giver output

2.5%	97.5%
-16.982860	-1.665015

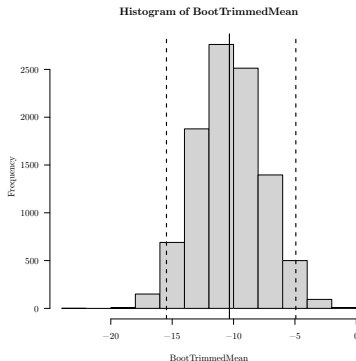
Bootstrap-metoden er i princippet meget fleksibel. I eksemplet nedenfor findes et 95%-percentil interval for *medianen*

```
n = 10
B = 1000
x <- rexp(n, rate = 1)
boot_medians <- replicate(B, median(sample(x, replace=TRUE)))
perc <- quantile(boot_medians, probs = c(0.025, 0.975))
```

Ved hjælp af et simulationseksperiment estimerer vi dækningssandsynligheden til 94% i dette tilfælde.

Bootstrap fordeling af 25%-trimmed mean og histogram.

```
for(i in 1:B) {  
  sampleILEC <- sample(ILEC,m,replace=TRUE)  
  sampleCLEC <- sample(CLEC,n,replace=TRUE)  
  BootTrimmedMean[i] <- mean(sampleILEC,trim=0.25) - mean(sampleCLEC,trim=0.25)  
}
```



Bootstrap metoden kan bruges til at estimere Bias $\hat{\theta} = \mathbb{E}[\hat{\theta}] - \theta$.

Bias vurderes bedst relativt til SE , $b = \text{Bias} / SE$

[Chihara and Hesterberg(2022)] advarer mod tab af nøjagtighed af percentil-intervaller, hvis $|b|$ er større end 0.02.

I [Chihara and Hesterberg(2022)] Verizon-eksempel med ratio-of-means fås $b = 0.2172$ så percentil-intervallet er nok ikke særligt nøjagtigt.

For trimmed-mean eksemplet fås:

```
> mean(BootTrimmedMean) - (mean(ILEC,trim=0.25) - mean(CLEC,trim=0.25))  
[1] 0.009031481  
> sd(BootTrimmedMean)  
[1] 2.688956
```

Her bør vi kunne have tillid til percentil-intervallet.

For et datasæt $x_1, \dots, x_n$ er der n^n mulige udfald $\mathbf{x}^* = (x_1^*, \dots, x_1^n)$ af den teoretiske bootstrap fordeling.

Når vi trækker N bootstrap samples approksimerer vi denne fordeling, og der derfor to kilder til variation:

- 1 Det oprindelige udfald $\mathbf{x}$ er udfald af den bagvedliggende s.v. $\mathbf{X}$.
- 2 N Bootstrap samples $\mathbf{x}^*$ er trukket ved hjælp af $\hat{F}_n$.

Vi kan (i princippet) gøre variationen fra (2) så lille som vi ønsker ved at gøre N stor. I §5.9 anbefales $N = 10000$.

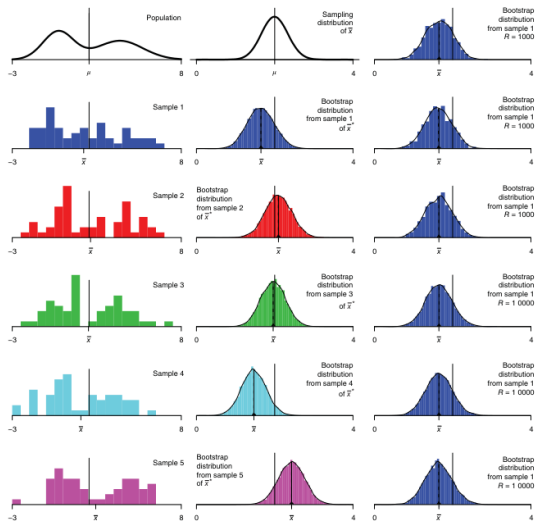


Figure 5.17 Bootstrap distribution for the mean, $n = 50$. The left column shows the population and five samples. The middle column shows the sampling distribution, and bootstrap distributions from each sample. The right column shows five more bootstrap distributions from the first sample, with $N = 1000$ or $N = 10^4$.

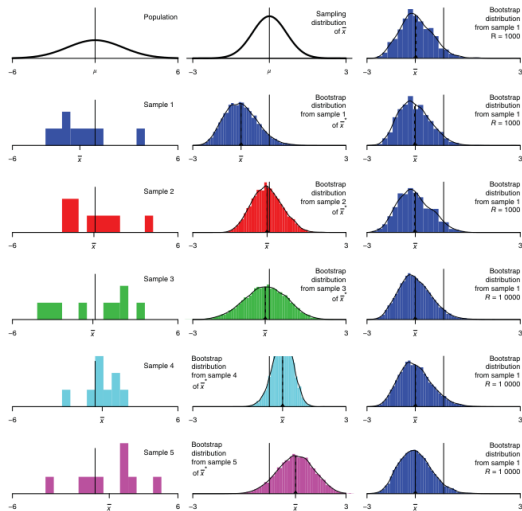


Figure 5.18 Bootstrap distributions for the mean, $n = 9$. The left column shows the population and five samples. The middle column shows the sampling distribution and bootstrap distributions from each sample. The right column shows five more bootstrap distributions from the first sample, with $N = 1000$ or $N = 10^4$.

Idéen i *bootstrap t confidence intervals* er at bootstrappe t -teststørrelsen:

$$t = \frac{\bar{X}_{\cdot} - \mu_0}{\sqrt{\frac{s^2(\mathbf{X})}{n}}},$$

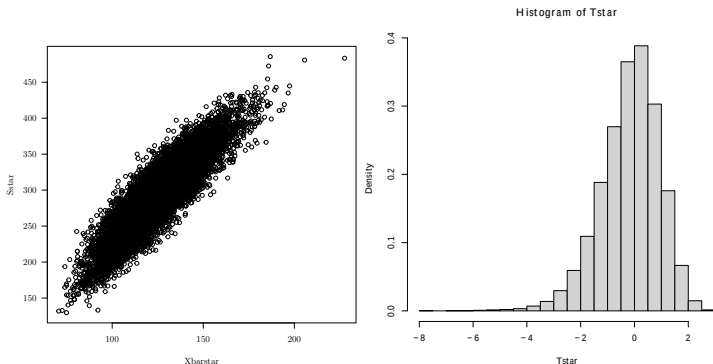
det vil sige finde bootstrap-fordelingen af

$$T^* = \frac{\bar{X}_{\cdot}^* - \bar{X}_{\cdot}}{s^*/\sqrt{n}}.$$

```
Arsenic <- Bangladesh$Arsenic
xbar <- mean(Arsenic)
N <- 10^4
n <- length(Arsenic)

Tstar <- numeric(N)
for (i in 1:N)
{
  xboot <- sample(Arsenic, size = n, replace = T)
  Tstar[i] <- (mean(xboot)-xbar) / (sd(xboot)/sqrt(n))
}
```

R-kode til arsenik-eksempel.



Scatterplot med s^* afsat mod $\bar{x}^*$ viser at disse er positivt korrelerede.

Bootstrap fordelingen af T bliver venstre skæv.

Det sædvanlige t -test udledes ved at benytte, at for én normalfordelt observationsrække har vi

$$1 - \alpha = \mathbb{P}(-t_{1-\alpha/2}(n-1) \leq T(\mathbf{x}) \leq t_{1-\alpha/2}(n-1))$$

Bruges samme udledning på

$$1 - \alpha = \mathbb{P}(Q_1 \leq T \leq Q_2)$$

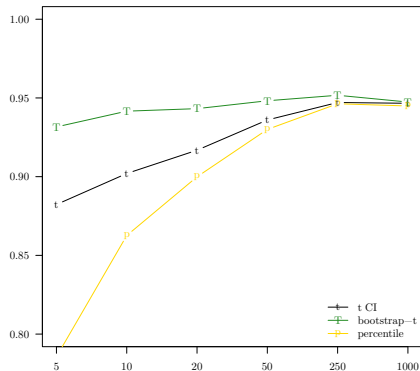
fås bootstrap bootstrap t confidence intervals:

Bootstrap t Confidence Interval for μ

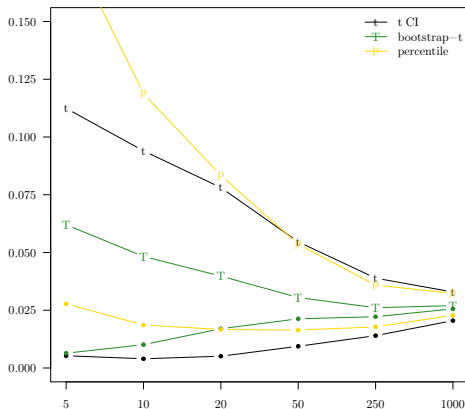
For each of many resamples, calculate the bootstrap t statistic $T^* = \frac{\bar{X} - \bar{x}}{S^*/\sqrt{n}}$.

Let $\hat{Q}_1$ and $\hat{Q}_2$ be the empirical $\alpha/2$ and $(1 - \alpha/2)$ quantiles of T^* , respectively (i.e. the interval between them contains the middle $(1 - \alpha) \times 100\%$ of all the bootstrap t statistics). The bootstrap t confidence interval is

$$\left(\bar{x} - \hat{Q}_2 \frac{s}{\sqrt{n}}, \bar{x} - \hat{Q}_1 \frac{s}{\sqrt{n}} \right).$$



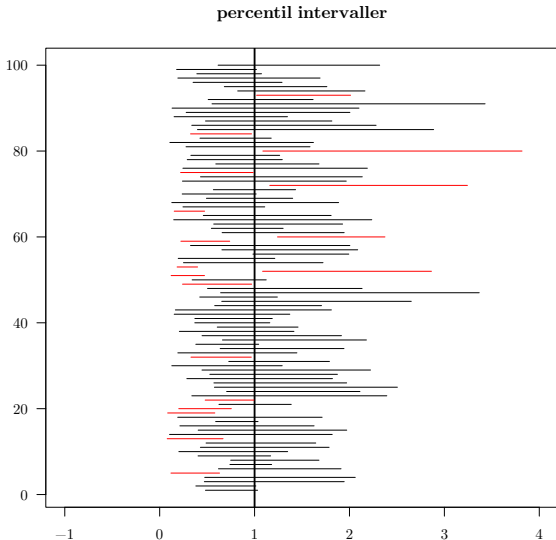
Estimerede dækningssandsynligheder tre slags konfidensintervaller for middelværdien som funktion af n . Data er simuleret fra uafhængige standard eksponential fordelte stokastiske variable.

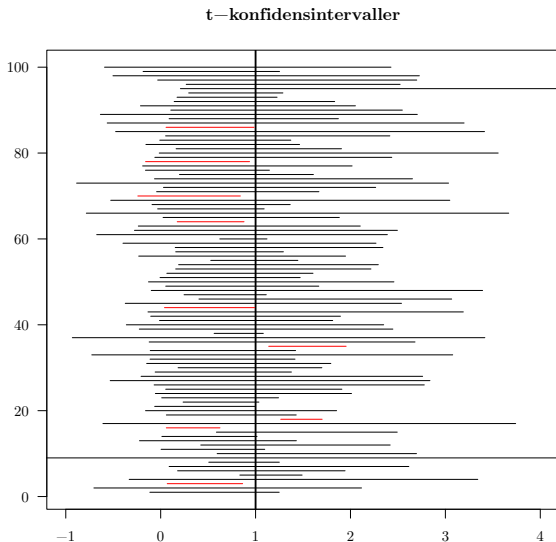


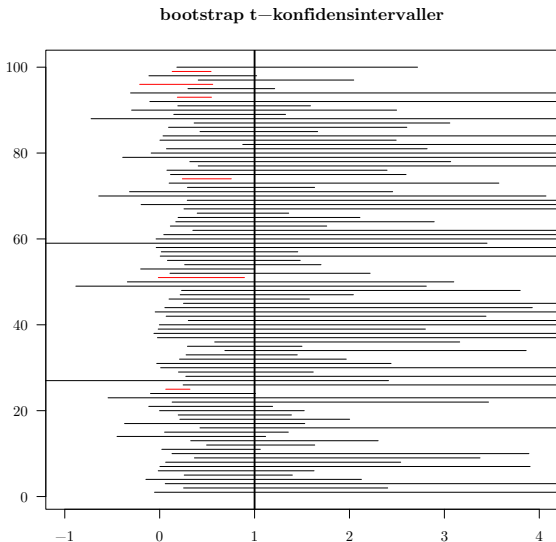
Figur svarende til Figure 7.14 i MSRR.

For $n = 5, 10, 20, 50, 250, 1000$ ses estimerede sandsynligheder for at konfidensintervallerne ligger over (nederste kurver) eller under (øverste kurver).

De næste slides viser 100 simulerede konfidensintervaller af hver type







Bibliografi



Laura M Chihara and Tim C Hesterberg.

Mathematical statistics with resampling and R (3rd edition).

John Wiley & Sons, 2022.